

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – VIII

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Class: TYIT	Batch: 1
Date of Assignment: 06-08-2026	Date/Time of Submission: 06-08-2026

a. Create an HTML page that would be the starting point of the application. This HTML will contain a link to the file named manifest.json. This is an important file that would be created in the next step.

Code:

Manifest.json:

```
{
  "name": "Student PWA",
  "short_name": "Student",
  "start_url": "/",
  "display": "standalone"
}
```

Add to _Layout.cshtml

Inside the <head> section:

```
<link rel="manifest" href="/manifest.json" />
```

Index.cshtml:

```
<h2> Student Record</h2>
```

```
<button onclick="loadStudent () ">Load
```

```
Student</button>
```

```
<p id="data"></p>
```

```
<script>
```

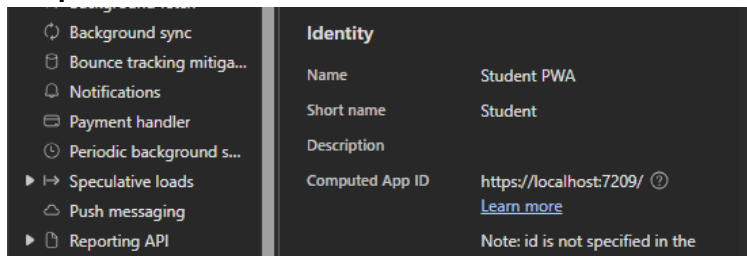
```
function loadStudent(){
  fetch("/api/student")
    .then (response => response.json ())
    .then (student=> {
```

```
document.getElementById("data").innerHTML=
  "Roll No : "+student.rollNo+
  "<br>Name : "+student.name;
  });
}
```

```
</script>
```

Step: run (open <https://localhost:7216>) and go to the console and click application and go to the manifest file

Output:



b. Create a simple Web API that returns Student data in JSON format.

Code:

student.cs

```
namespace StudentPWA.Models
{
    public class Student
    {
        public int RollNo { get; set; }
        public string Name { get; set; }
    }
}
```

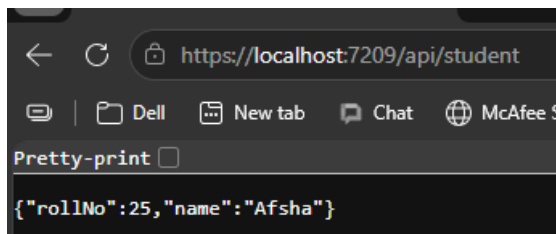
StudentController.cs

```
using Microsoft.AspNetCore.Mvc;
```

```
namespace StudentPWA.Controllers
```

```
{
    [Route("api/[controller]")]
    [ApiController]
    public class StudentController : ControllerBase
    {
        [HttpGet]
        public IActionResult Get ()
        {
            return Ok(new
            {
                RollNo = 25,
                Name = "Afsha"
            });
        }
    }
}
```

Step: run (open <https://localhost:7163>) its your portal no. and add this (/api/student)



StudentPWA Home Privacy

Student Record

Load Student

Roll No : 25

Name : Afsha

Offline page(optional:)

offline.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Offline</title>
</head>
<body>
```

```

<main>
  <h2> You are Offline</h2>
  <p>Please check your internet connection.</p>
  <button onclick="location.reload()">Try Again</button>
</main>

</body>
</html>

```

Create `wwwroot-service-worker.js`:

```

const CACHE = "v1";
self.addEventListener("install", e => {
  e.waitUntil(
    caches.open(CACHE).then(cache => {
      return cache.addAll([
        "/",
        "/offline.html"
      ]);
    });
  });

console.log("Service Worker Installed");
});
self.addEventListener("fetch", e => {
  e.respondWith(
    fetch(e.request).catch(() => caches.match("/offline.html"))
  );
});

```

Create `_Layout.cshtml`:

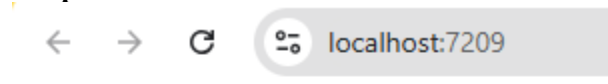
Before the closing `</body>` tag:

```

<script>
  if ('serviceWorker' in navigator) {
    navigator.serviceWorker.register('/service-worker.js');
  }
</script>

```

Output:



You are Offline

Please check your internet connection.

Try Again